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Compressed air station

Abstract

The application relates to a compressed air station comprising at least two compressed air components that yield waste heat, wherein each compressed air component is designed either as a compressor, in particular as a screw compressor, or as a refrigeration dryer, wherein at least one of the compressed air components, namely a refrigeration dryer (12), is connected to the exhaust air duct (13), and wherein a further compressed air component is connected to the same exhaust air duct (13), wherein a compressed air refrigeration heat exchanger (23) is provided within the refrigeration dryer (12), in which the compressed air is cooled by way of a refrigerant conducted in a refrigerant circuit (24), wherein the refrigerant circuit (24) comprises a refrigerant compressor (25), a condenser (26), an expansion valve (27) and the compressed air refrigeration heat exchanger (23), wherein the compressed air station further comprises a dryer exhaust air duct (15), which is provided for discharging a cooling air flow that is conducted through the refrigeration dryer (12), and which connects a cooling air outlet (19) of the refrigeration dryer (12) to a refrigeration dryer connection (16) on the exhaust air duct (13), wherein the refrigeration dryer (12) has a fan (20) with a speed-adjustable fan motor (21), and the fan (20) is designed to convey the cooling air flow even against a backpressure currently prevailing in the exhaust air duct (13), wherein the refrigeration dryer (12) has a flow sensor (30) for detecting a respective current value for the cooling air volume flow $V_{sub.act}$, and wherein the refrigeration dryer (12) has a controller (22) or interacts with a controller (22), which is configured and designed to record and process the data from the flow sensor (30) and to actuate the fan motor (21) of the fan (20) in such a way that, regardless of the current backpressure in the exhaust air duct (13), the respective current cooling air volume flow $V_{sub.act}$ follows a setpoint for the cooling air volume flow $V_{sub.soll}$.

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Background/Summary

(1) The invention relates to a compressed air station comprising at least two compressed air components that yield waste heat, wherein each compressed air component is designed either as a compressor, in particular as a screw compressor, or as a refrigeration dryer, and at least one exhaust air duct for discharging waste heat from a room.

(2) A compressed air refrigeration heat exchanger is provided within the refrigeration dryer, in which the compressed air is cooled by way of a refrigerant conducted in a refrigerant circuit, wherein the refrigerant circuit comprises a refrigerant compressor, a condenser, an expansion valve and the compressed air refrigeration heat exchanger. The waste heat produced by the compressor(s) is traditionally already discharged via one or more exhaust air ducts from a plant room in which the compressed air components are installed. In the case of refrigeration dryers, which usually produce a small amount of waste heat compared to the compressors present within a compressed air station, the waste heat has traditionally been either introduced into the plant room or conducted away via extraction hoods arranged above the refrigeration dryers. However, particularly when discharging the waste heat from the refrigeration dryer(s) into the plant room or even when this is conducted away only incompletely via an extraction hood arranged above the refrigeration dryers, the problem arises that the ambient air in the plant room heats up and the efficiency both of the compressors, in particular the screw compressors, and of the refrigeration dryers decreases.

(3) In contrast, the object of the present invention is to propose a compressed air station and a corresponding method in which an improved discharging of waste heat from a plant room is made possible, including the waste heat from a refrigeration dryer present in the compressed air station.

(4) This object is achieved in terms of the device by a compressed air station having the features of claim 1 and in terms of the method by a method for actuating the fan motor of a fan of a refrigeration dryer according to the features of claim 12. Advantageous further developments are specified in the dependent claims.

(5) In terms of the device, the compressed air station is characterized in that it further comprises a dryer exhaust air duct, which is provided for discharging a cooling air flow that is conducted through the refrigeration dryer, and which connects a cooling air outlet of the refrigeration dryer to a refrigeration dryer connection on the exhaust air duct, wherein the refrigeration dryer has a fan with a speed-adjustable fan motor, and the fan is designed to convey the cooling air flow even against a backpressure currently prevailing in the exhaust air duct, wherein the refrigeration dryer has a flow sensor for detecting a respective current value for the cooling air volume flow $V_{sub.act}$,

and wherein the refrigeration dryer has a controller or interacts with a controller, which is configured and designed to record and process the data from the flow sensor and to actuate the fan motor of the fan in such a way that, regardless of the current backpressure in the exhaust air duct, the respective current cooling air volume flow $V_{sub.act}$ follows a setpoint for the cooling air volume flow $V_{sub.soll}$.

(6) In terms of the method, it is proposed that the fan of a refrigeration dryer within a compressed air station is actuated in order to compensate for even a fluctuating backpressure within an exhaust air duct, which is also fed by at least one further compressed air component, the method being designed as follows:

(7) A method for actuating the fan motor of a fan of a refrigeration dryer within a compressed air station, wherein the compressed air station comprises at least two compressed air components that yield waste heat, wherein each compressed air component is designed either as a compressor, in particular as a screw compressor, or as a refrigeration dryer, and at least one exhaust air duct for discharging waste heat from a room, wherein at least one of the compressed air components, namely a refrigeration dryer, is connected to the exhaust air duct, and wherein a further compressed air component is connected to the same exhaust air duct,

(8) wherein a compressed air refrigeration heat exchanger is provided within the refrigeration dryer, in which the compressed air is cooled by way of a refrigerant conducted in a refrigerant circuit, wherein the refrigerant circuit comprises a refrigerant compressor, a condenser, an expansion valve and the compressed air refrigeration heat exchanger,

(9) wherein the compressed air station further comprises a dryer exhaust air duct, which is provided for discharging a cooling air flow that is conducted through the refrigeration dryer, and which connects a cooling air outlet of the refrigeration dryer to a refrigeration dryer connection on the exhaust air duct,

(10) wherein the refrigeration dryer has a fan with a speed-adjustable fan motor, and the fan is designed to convey the cooling air flow even against a backpressure currently prevailing in the exhaust air duct, wherein the method comprises the following steps: specifying a setpoint $V_{sub.soll}$ for the cooling air volume flow, detecting a respective current value for the cooling air volume flow $V_{sub.act}$, and actuating the fan motor (21) of the fan (20) in such a way that the respective current cooling air volume flow $V_{sub.act}$ follows the respectively specified setpoint for the cooling air volume flow $V_{sub.soll}$.

(11) One core consideration of the present invention is that of controlling the fan motor of the fan in the refrigeration dryer in such a way that a residual pressure of the exhaust air flow from the refrigeration dryer is adjusted to the pressure level of an exhaust air duct system used jointly with at least one screw compressor. The speed of the fan motor of the fan is therefore varied such that the amount of exhaust air from the refrigeration dryer per unit of time, i.e. the cooling air volume flow of the refrigeration dryer, remains unchanged regardless of the current backpressure in the exhaust air duct.

(12) The condensation pressure and the cooling capacity of the refrigeration dryer can thus be kept at the same level regardless of the current backpressure, wherein it remains possible for the condensation pressure or the cooling capacity to vary according to other criteria, but the current backpressure in the exhaust air duct or in the exhaust air duct system has no influence on the condensation pressure or the cooling capacity of the refrigeration dryer. A current residual pressure will be understood here to mean the current pressure reserve of the fan for overcoming additional flow resistances, such as that of an exhaust air duct for example. The maximum achievable residual pressure of the fan used in the refrigeration dryer is to be regarded as sufficient if the backpressure, as results from the interaction with other components, such as a connected screw compressor, and the additional flow resistances, as occur inter alia in an exhaust air duct of customary length, can be overcome.

(13) Although, with regard to the refrigerant circuit, mention is made of a condenser on the one

hand and an expansion valve on the other hand, it is clarified that the refrigerant circuit need not necessarily be operated with a phase transition of gas-liquid liquid-gas, but rather in the case of certain refrigerants is also operated via transcritical processes, such as for example in the case of CO₂ (R-744). There is then no liquefaction, but rather heat is output on the high-pressure side of the refrigerant circuit and the gas phase is retained. The gas cooler outlet temperature is then to be seen as a value equivalent to the condensation temperature.

(14) A speed-adjustable fan motor in the sense of the invention will be understood to be a fan motor that can be adjusted with regard to its speed, in particular by frequency reversal or by phase control.

(15) In the present application, the cooling air flow conducted through the refrigeration dryer is referred to as the cooling air flow per se. Where the cooling air volume flow is mentioned, this refers to the quantitative value as a volume flow, for example expressed in the unit m³/s, i.e. the volume of cooling air transported by the cooling air flow per period of time.

(16) In a preferred embodiment of the present invention, the flow sensor is designed as a differential pressure sensor.

(17) In a preferred embodiment of the compressed air station proposed here, the fan of the refrigeration dryer is thus also designed as a radial fan. Admittedly, traditionally used axial fans have a lower power consumption at the nominal point and thus initially appear more favourable from an energy point of view. However, this applies particularly in connection with traditionally used refrigeration dryers, which blow out their waste heat directly into a plant room in which they are installed. Radial fans appear more suitable for direct connection to an exhaust air duct or to an exhaust air duct system since they are able to ensure a higher residual pressure.

(18) In a further preferred embodiment, it is provided that a screw compressor and the refrigeration dryer are connected to one another via a compressed air line, and the compressed air line is designed to transfer the compressed air output by the screw compressor to the refrigeration dryer for drying purposes, wherein a cooling air outlet of the screw compressor is connected via a compressor connection to the exhaust air duct, to which the refrigeration dryer is also connected. In this possible embodiment, the screw compressor and the refrigeration dryer convey their waste heat via corresponding cooling air outlets to the same exhaust air duct. At the same time, the compressed air output by the screw compressor is transferred to the refrigeration dryer for drying purposes. However, it is also conceivable that, although the compressed air generated by the screw compressor is transferred to the refrigeration dryer, the screw compressor and the refrigeration dryer output their waste heat to different exhaust air ducts.

(19) In the compressed air station proposed here, one or more oil-injected screw compressors and/or one or more oil-free compressors may be used.

(20) In a preferred embodiment of the compressed air station proposed here, the controller interacts with an ambient air sensor, preferably a temperature sensor, for detecting a value representative of the state of the supply air, in particular a value representative of the supply air temperature, and transmitting it to the controller.

(21) In a possible embodiment of the present invention, two or more screw compressors and/or two or more refrigeration dryers are connected to a common exhaust air duct. Particularly large pressure fluctuations or pressure differences may occur in a common exhaust air duct used in this way, and therefore an adjustment of the fan motor of the fan of the refrigeration dryer, as proposed in the context of the present invention, is particularly important.

(22) In a preferred embodiment, the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$ results from a design value for the cooling air volume flow $V_{\text{sub.nenn}}$, which is specific to the given refrigerant circuit with the given refrigerant compressor, according to the following formula:

$$V_{\text{sub.soll}} = V_{\text{sub.nenn}} \cdot F_{\text{sub.Hub/KMK}} \cdot F_{\text{sub.T}}$$

(23) where the factors $F_{\text{sub.Hub/KMK}}$ and $F_{\text{sub.T}}$ are correction factors, each of which can alternatively be set to 1 or to a value between 0 and 1.

(24) In a preferred embodiment, an exhaust air damper is arranged in the dryer exhaust air duct, which damper is designed and configured to close the dryer exhaust air duct when the refrigerant compressor is idle. This prevents the exhaust air from flowing back into a plant room in which the refrigeration dryer is installed.

(25) In this case, in a preferred embodiment, the exhaust air damper may be designed as a gravity-operated exhaust air damper, which opens whenever the fan is conveying the exhaust air through the dryer exhaust air duct and closes whenever the fan is idle.

(26) In an alternatively possible embodiment, the exhaust air damper may interact with a drive motor, wherein the drive motor is actuated by the controller in such a way that the exhaust air damper is opened or closed depending on the operating state of the fan. In particular, the exhaust air damper is or will be opened when the fan is operating. The exhaust air damper is or will be closed in particular when the fan is idle.

(27) In a specific further development of the proposed method, the respective current value for the cooling air volume flow $V_{\text{sub.act}}$ is detected by way of a differential pressure measurement. A differential pressure measurement is a relatively simple and at the same time reliable measurement method for determining with sufficient certainty a current value for the cooling air volume flow $V_{\text{sub.act}}$.

(28) In a preferred embodiment, the preferred differential pressure measurement is performed immediately before the cooling air flow flows into the fan. In other words, the pressure is detected in the region immediately before the air flow flows into the fan and is compared with a reference pressure. Such a reference pressure may be provided for example between the condenser and the fan. As already mentioned above, this is a particularly favourable position within the course of the cooling air flow for performing a differential pressure measurement.

(29) In a possible embodiment, the setpoint for the cooling air volume flow may be adapted as a function of the load state of the refrigerant compressor and/or as a function of the ambient temperature. If the cooling air volume flow is adapted as a function of the load state of the refrigerant compressor, in the case of just one refrigerant compressor for example the usage state of one, possibly frequency-controlled, refrigerant compressor is to be taken into account. If a plurality of refrigerant circuits are operated in parallel with one another in a plant, the load state may also include for example the extent to which each refrigerant compressor is loaded.

(30) In a preferred embodiment of the present method, the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$ results from a design value for the cooling air volume flow $V_{\text{sub.nenn}}$, which is specified for the refrigerant circuit and the refrigerant compressors installed therein.

(31) In a further preferred exemplary embodiment, the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$ results from a design value for the cooling air volume flow $V_{\text{sub.nenn}}$, which is specified for the refrigerant circuit and the refrigerant compressors installed therein, taking into account one or more correction factors F .

(32) In a specifically preferred embodiment of the present method, the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$ results from the design-based nominal value for the cooling air volume flow $V_{\text{sub.nenn}}$, according to the following formula:

$V_{\text{sub.soll}} = V_{\text{sub.nenn}} \cdot F_{\text{sub.Hub/KMK}} \cdot F_{\text{sub.T}}$, where $F_{\text{sub.Hub/KMK}}$ is a correction factor for taking into account the respective current stroke volume in the refrigerant circuit, and $F_{\text{sub.T}}$ is a correction factor for taking into account temperature fluctuations in the supply air of the cooling air flow, where: $0 \leq F_{\text{sub.Hub/KMK}} \leq 1$ and $0 \leq F_{\text{sub.T}} \leq 1$. Therefore, $V_{\text{sub.soll}}$ results directly from $V_{\text{sub.nenn}}$, with one or more correction factors being used in certain operating situations and/or under certain operating conditions, which correction factors ensure that $V_{\text{sub.soll}}$ is reduced in comparison to $V_{\text{sub.nenn}}$. $V_{\text{sub.soll}}$ can thus be a certain percentage between 0 and 100% of $V_{\text{sub.nenn}}$.

(33) In a further preferred embodiment, a correction factor $F_{\text{sub.T}}$ takes into account a current value of the supply air temperature, wherein the value $F_{\text{sub.T}}$ is set to 1 in the case of supply air

temperatures above a limit temperature $T_{\text{sub.0,amb}}$, and $0 \leq F_{\text{sub.T}} < 1$ applies only in the case of cooling air inlet temperature values $T < T_{\text{sub.0,amb}}$. In a very specifically preferred embodiment, the correction factor $F_{\text{sub.T}}$ for the temperature T of the supply air (cooling air inlet temperature) in ranges below the limit temperature $T_{\text{sub.0,amb}}$ can be calculated as follows:

$$(34) T_{0, \text{amb}} = \frac{T}{T_{0, \text{amb}} - T_{\text{amb, act}}},$$

(35) where $\Delta_{\text{sub.T}}$ denotes a dryer-specific supplement in ° C., $T_{\text{sub.0,amb}}$ denotes a fixed limit temperature, and $T_{\text{sub.amb,act}}$ denotes the current supply air temperature.

(36) In a further preferred embodiment, the correction factor $F_{\text{sub.Hub/KMK}}$ can be formed from a ratio of the current stroke volume and of the maximum stroke volume in the refrigerant circuit, according to the following formula

$$(37) F_{\text{Hub / KMK}} = \frac{\text{currentstrokevolumeKMK}}{\text{maximumstrokevolumeKMK}}$$

(38) By way of the correction factor $F_{\text{sub.Hub/KMK}}$, therefore, it is possible to take into account operating situations in which the current stroke volume KMK is reduced in comparison to a maximum stroke volume KMK for which the refrigerant circuit is designed. This may be achieved for example in that, if using multiple refrigerant compressors, one or more fluid compressors are switched off or, if using one refrigerant compressor with an adjustable output, the output of the refrigerant compressor is currently reduced.

(39) In a specifically preferred embodiment of the method according to the invention, a PID controller, a PI controller, a deadband controller or a three-point controller may be used to actuate the fan motor of the fan in order to bring the current cooling air flow $V_{\text{sub.act}}$ towards the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$, wherein a control deviation e results from $V_{\text{sub.soll}} - V_{\text{sub.act}}$, and wherein $V_{\text{sub.soll}}$ denotes the setpoint for the cooling air volume flow and $V_{\text{sub.act}}$ denotes the current cooling air volume flow.

Description

(1) The invention will be explained in greater detail below, including with regard to further features and advantages, based on the description of exemplary embodiments and with reference to the following drawings, in which:

(2) FIG. 1 shows an exemplary embodiment of a compressed air station according to the invention, in a schematic illustration;

(3) FIG. 2 shows an embodiment of a compressed air station according to the invention, modified in comparison to the embodiment shown in FIG. 1, in a schematic illustration;

(4) FIG. 3 shows an embodiment of a compressed air station, once again modified in comparison to FIGS. 1 and 2, in a schematic illustration;

(5) FIG. 4 shows an embodiment of a compressed air station, once again modified in comparison to FIGS. 1 to 3, in a schematic illustration;

(6) FIG. 5 shows an embodiment of a compressed air station, once again modified in comparison to FIGS. 1 to 4, in a schematic illustration;

(7) FIG. 6 shows a schematic illustration to explain the mode of operation of a refrigeration dryer according to the present invention;

(8) FIG. 7 shows a flowchart to explain an embodiment of the method according to the invention; and

(9) FIG. 8 shows an embodiment of a possible curve of the correction factor $F_{\text{sub.T}}$ as a function of the supply air of the cooling air flow $T_{\text{sub.amb}}$.

(10) FIG. 1 shows an exemplary embodiment of a compressed air station according to the invention, in which, as compressed air components, a screw compressor **11** and a refrigeration dryer **12** are connected to a common exhaust air duct **13**. The screw compressor **11** supplies compressed air via a compressed air line **14** to the refrigeration dryer **12**. Via a further compressed

air line **43**, the compressed air dried in the refrigeration dryer **12** is supplied to a consumer **44**. Instead of one consumer, a plurality, in particular a large number, of consumers may also be supplied with compressed air through the further compressed air line **43** via a compressed air network.

(11) The screw compressor **11** and the refrigeration dryer **12** produce waste heat, which according to the invention is discharged via the common exhaust air duct **13** by way of corresponding cooling air flows. A screw compressor exhaust air duct **45** forms the first section of a common exhaust air duct **13**, to which the refrigeration dryer **12** is also connected further downstream via a dryer exhaust air duct **15**. Specifically, the screw compressor **11** has a cooling air outlet **18**, to which the screw compressor exhaust air duct **45** is directly connected. The refrigeration dryer **12** likewise has a cooling air outlet **19**, to which the dryer exhaust air duct **15** is directly connected, namely preferably in such a way that only the cooling air or the waste heat of the refrigeration dryer **12** is discharged and no mixing takes place with ambient air not conducted via the refrigeration dryer **12**.

(12) The dryer exhaust air duct **15** is connected to the common exhaust air duct **13** at the aforementioned refrigeration dryer connection **16**. In addition, an exhaust air damper **29** is arranged between the refrigeration dryer **12** and the refrigeration dryer connection **16** on the exhaust air duct **13**, in particular within the dryer exhaust air duct **15**, by means of which damper the dryer exhaust air duct **15** can be closed.

(13) A further exhaust air damper **46** may also be arranged between the screw compressor **11**, i.e. between the cooling air outlet **18** of the screw compressor **11** and the refrigeration dryer connection **16**, in particular within the screw compressor exhaust air duct **45**. As passive backflow dampers, the exhaust air damper **29** of the refrigeration dryer and/or the exhaust air damper **46** of the screw compressor can either be opened using a sufficient flow in the conveying direction or closed under the effect of gravity. However, it is also possible (see illustration in FIG. 6) to open and close the exhaust air damper **29** by means of a drive motor **39**. Of course, the exhaust air damper **46** of the screw compressor can also be opened and closed by means of a drive motor (not shown).

(14) FIG. 2 shows an embodiment of a compressed air station according to the invention, modified in comparison to the embodiment shown in FIG. 1, in which, as compressed air components, a screw compressor **11** and a first refrigeration dryer **12** and also a further refrigeration dryer **12'** are connected to a common exhaust air duct **13**. In this embodiment, the compressed air transferred from the screw compressor to the compressed air line **14** is split at a branching point **47** into a first partial line **48** and a second partial line **49**.

(15) The first refrigeration dryer **12** is arranged in the first partial line **48**, and the second refrigeration dryer **12'** is arranged in the second partial line **49**. The dried compressed air leaves the first refrigeration dryer **12** via a third partial line **50**. The dried compressed air that flows via the second partial line **49** into the second refrigeration dryer **12'** and is dried therein leaves the refrigeration dryer **12'** via a fourth partial line **51**. The third partial line **50** and the fourth partial line **51** are brought together at a joining point **52** and merge into the compressed air line **43**, which conducts the compressed air to at least one consumer **44**.

(16) The screw compressor **11**, the refrigeration dryer **12** and the refrigeration dryer **12'** each direct their cooling air flows into a common exhaust air duct **13**. To this end, the screw compressor **11** is connected to the exhaust air duct **13** in the manner already described with reference to the embodiment shown in FIG. 1. The first refrigeration dryer **12** is also connected to the common exhaust air duct **13**, in the manner already described with reference to the embodiment shown in FIG. 1, in order to transfer the cooling air or waste heat output from the cooling air outlet **19**. Downstream of the refrigeration dryer connection **16**, which is assigned to the first refrigeration dryer **12**, a second refrigeration dryer connection **53** is provided on the exhaust air duct **13**, at which the cooling air of the second refrigeration dryer **12'** is also introduced downstream into the exhaust air duct **13**. To this end, a cooling air outlet **54** of the second refrigeration dryer is connected to a dryer exhaust air duct **55**, which connects the cooling air outlet **54** of the second

refrigeration dryer **12'** to the refrigeration dryer connection **53** on the exhaust air duct **13**, so that the cooling air also of the second refrigeration dryer is discharged without any ambient air not conducted via the second refrigeration dryer **12'**. Also in this dryer exhaust air duct assigned to the second refrigeration dryer **12'**, an exhaust air damper **64** is provided so as to be able to close the exhaust air duct **55**, in particular when the second refrigeration dryer **12'** is idle.

(17) FIG. 3 shows an embodiment of a compressed air station, once again modified in comparison to the embodiments shown in FIG. 1 and FIG. 2, in which, as compressed air components, a first screw compressor **11**, a second screw compressor **11'**, a first refrigeration dryer **12** and a second refrigeration dryer **12'** are connected to a common exhaust air duct **13**. In the embodiment of a compressed air station illustrated in FIG. 3, two screw compressors are provided, namely the first screw compressor **11** and the second screw compressor **11'**, which operate in parallel with one another with regard to compressed air generation, i.e. the screw compressor **11** outputs compressed air on a first output line **56** and the screw compressor **11'** outputs compressed air on a second output line **57**. At a joining point **58**, the first output line **56** and the second output line **57** join to form a common compressed air line **14**. From the compressed air line **14**, the compressed air is conducted at a branching point **47** to a first partial line **48** and to a second partial line **49**, in which a respective refrigeration dryer **12** or **12'** is connected. The compressed air is thus dried by the two refrigeration dryers **12**, **12'** in parallel, so that the arrangement of the two refrigeration dryers corresponds exactly to the arrangement of the two refrigeration dryers according to the embodiment shown in FIG. 2. The discharging of the cooling air or waste heat via dryer exhaust air duct **15** or dryer exhaust air duct **55** also takes place exactly as in the arrangement shown in FIG. 2.

(18) In manner differing from the arrangement shown in FIG. 2, however, in the embodiment shown in FIG. 3 not only are two refrigeration dryers **12**, **12'** connected to the common exhaust air duct **13**, but also the aforementioned two screw compressors **11**, **11'**. The first screw compressor **11** is connected to the common exhaust air duct **13** in the manner already described with reference to FIGS. 1 and 2 and represents the component yielding waste heat that is arranged furthest upstream in relation to the flow direction of the exhaust air duct **13**. The screw compressor **11** is therefore the waste heat supplier placed most upstream within the common exhaust air duct **13**.

(19) The aforementioned second screw compressor **11'** has a cooling air outlet **59**, by which it is connected to a screw compressor exhaust air duct **60**. The screw compressor exhaust air duct **60** connects the cooling air outlet **59** of the second screw compressor **11'** to a compressor connection **61**, at which the screw compressor exhaust air duct **60** is connected to the common exhaust air duct **13**, namely at a section between the compressor connection **17** of the first screw compressor **11** and the refrigeration dryer connection **16** of the first refrigeration dryer **12** or the second refrigeration dryer connection **53** of the refrigeration dryer **12'**.

(20) FIG. 4 illustrates an embodiment that has once again been modified, in which, as compressed air components, a first refrigeration dryer **12** and a second refrigeration dryer **12'** are connected to a common exhaust air duct **13**. In this case, therefore, a common exhaust air duct **13** is not fed by one refrigeration dryer **12** and one screw compressor **11**, but rather by two refrigeration dryers **12**, **12'**. The connection of the refrigeration dryers **12**, **12'** corresponds to the connection of the refrigeration dryers **12**, **12'** in the arrangement of the embodiment shown in FIG. 3, with the sole exception that no cooling air or waste heat from a screw compressor **11**, **11'** is introduced upstream of the connection points where the refrigeration dryers **12**, **12'** are connected to the common exhaust air duct **13**.

(21) FIG. 5 illustrates an embodiment that has once again been modified, in which, as compressed air components, a first screw compressor **11** discharges waste heat into a first exhaust air duct **13** via a screw compressor exhaust air duct **45**. The screw compressor **11** supplies compressed air via a compressed air line **14** to a first refrigeration dryer **12**. The compressed air dried in the refrigeration dryer **12** is conducted via a first output line **66** to a joining point **68**, to which compressed air dried in a second refrigeration dryer **12'** is also conducted via a second output line **67**. From the joining

point **68**, the combined dried compressed air is fed to a consumer **44**.

(22) A second screw compressor **11'** generates compressed air and transfers this compressed air via the compressed air line **14'** to the aforementioned second refrigeration dryer **12'**. Waste heat from the second screw compressor **11'** is supplied via a screw compressor exhaust air duct **60** to a second exhaust air duct **13'**, which is separate from the exhaust air duct **13**. Waste heat from the first refrigeration dryer **12** is also discharged into this second exhaust air duct **13'** via a dryer exhaust air duct **15**. The second refrigeration dryer **12'**, which in terms of compressed air is connected to the second screw compressor **11'**, in contrast feeds its exhaust air via a dryer exhaust air duct **55** to the exhaust air duct **13**, which is also fed by the first screw compressor **11**. Here, therefore, the pairs of screw compressors and refrigeration dryers connected one behind the other are connected to two different exhaust air ducts **13**, **13'** in a crossed fashion.

(23) The basic structure and the basic mode of operation of a refrigeration dryer **12**, **12'** according to the present invention will be explained in greater detail below with reference to FIG. **6**. The refrigeration dryer first has a compressed air inlet **62** and a compressed air outlet **63**. The compressed air flowing into the refrigeration dryer **12**, **12'** via the compressed air inlet **62** is cooled by a refrigerant at a compressed air refrigeration heat exchanger **23** and leaves the refrigeration dryer through the compressed air outlet **63**. As is well known to a person skilled in the art, prior to flowing into the compressed air refrigeration heat exchanger **23**, the compressed air is usually pre-cooled in a pre-heat exchanger, namely by the compressed air that has already flowed through the compressed air refrigeration heat exchanger **23**, which is thus heated again before flowing out of the compressed air outlet **63**. However, the pre-heat exchanger described above is not shown in the present case.

(24) To provide the cooling capacity required at the compressed air refrigeration heat exchanger **23**, the latter is part of a refrigerant circuit **24**, which forms a compression refrigeration machine, known per se, and thus comprises, as seen in the direction of flow of the refrigerant from the compressed air refrigerant heat exchanger **23** a refrigerant compressor **25**, a condenser **26** connected thereto and an expansion valve **27** downstream thereof. The gas expanded after flowing through the expansion valve **27** is cooled by the expansion process and in the compressed air refrigerant heat exchanger **23** transfers cold to the compressed air. The refrigerant compressor **25** compresses the refrigerant. The heat produced during this is output in the condenser **26** to the supply air. For this purpose, a cooling air flow **65** is conducted through the refrigeration dryer **12**, **12'**, the supply air usually being ambient air and being sucked into the refrigeration dryer **12**, **12'** at a supply air inlet **41**, namely under the effect of a fan **20** driven by a fan motor **21**.

(25) The cooling air flow **65** absorbs heat in the condenser **26**, so that the refrigerant in the condenser **26** is cooled. According to the invention, the resulting waste heat is to be introduced via a dryer exhaust air duct **15** to an exhaust air duct **13**, into which waste heat from other compressed air-generating or compressed air-processing components is also conducted. For this purpose, the dryer exhaust air duct **15** is connected directly to the cooling air outlet **19** of the refrigeration dryer **12**, **12'**. An exhaust air damper **29**, which can be moved by way of a drive motor **39**, in particular from an open position to a closed position or vice versa, interacts with a controller **22**, which actuates the exhaust air damper **29** and closes or opens it as required.

(26) However, the controller **22** also controls the fan motor **21** of the fan **20**.

(27) The fan motor **21** is speed-adjustable. A current value for the cooling air volume flow $V_{\text{sub.act}}$ is detected within the cooling air flow **65** by a flow sensor **30**. This preferably takes place in that the flow sensor **30** is designed as a differential pressure sensor and detects on the one hand the pressure within the cooling air flow **65** in the region of an inflow nozzle **31**, which is arranged immediately in front of the fan **20**, and on the other hand in an upstream region of the cooling air flow way before it flows into the inflow nozzle **31**. From these two pressure values and the resulting pressure difference, it is possible to calculate the flow rate of the cooling air flow and thus, based on the given flow cross-section, the current volume flow. The current value of the

cooling air volume flow $V_{sub.act}$ detected by the flow sensor **30** is transmitted to the controller **22**, which then actuates the fan motor **21** in such a way that, regardless of the current backpressure in the exhaust air duct **13**, a respectively specified value for the cooling air volume flow $V_{sub.soll}$ is adhered to as precisely as possible, that is to say the fan **20** is adjusted in such a way that the current cooling air volume flow $V_{sub.act}$ follows the respectively pre-set setpoint for the cooling air volume flow $V_{sub.soll}$.

(28) Although the differential pressure measurement proposed here is a particularly simple, inexpensive and reliable method for determining the current cooling air volume flow $V_{sub.act}$, the invention is of course not limited to this specific way of detecting the value of the cooling air volume flow. Instead, various other measurement techniques and methods are conceivable for detecting the current value of the cooling air volume flow.

(29) A preferred adjustment, taking into account the respective current value for the cooling air volume flow $V_{sub.act}$, will be explained in greater detail below with reference to the flowchart shown in FIG. 7:

(30) In a step **100**, the refrigeration dryer **12**, **12'** is first started. In a step **101**, the cooling capacity requirement is determined and the refrigerant circuit **24** is started by setting the refrigerant compressor **25** in operation. In step **102**, the fan **20** is started at a specified minimum speed.

(31) Then, in step **103**, a current differential pressure $\Delta p_{sub.act}$ is determined by the flow sensor **30**, and in a step **104** a current cooling air volume flow $V_{sub.act}$ is calculated from the current differential pressure $\Delta p_{sub.act}$. In parallel with this, a setpoint for the cooling air volume flow $V_{sub.soll}$ is defined. In the present embodiment, this takes place by first reading in a signal for the supply air temperature $T_{sub.amb}$ in a step **105**. In a step **106**, a setpoint for the cooling air volume flow $V_{sub.soll}$ is calculated, taking into account the temperature of the supply air $T_{sub.amb}$ and optionally also a current stroke volume. Based on a design value for the cooling air volume flow $V_{sub.nenn}$, which is specific to the given refrigerant circuit **24** with the given refrigerant compressor **25**, a setpoint for the cooling air volume flow is determined according to the following formula, using two correction factors:

$$V_{sub.soll} = V_{sub.nenn} \cdot F_{sub.Hub} / KMK \cdot F_{sub.T},$$

(32) where the factors $F_{sub.Hub} / KMK$ and $F_{sub.T}$ are correction factors. The correction factor $F_{sub.T}$ is used only in the case where the supply air temperature is below a limit temperature $T_{sub.0,amb}$, and is otherwise set equal to 1, cf. in this regard FIG. 8, which shows an example of the curve of the correction factor $F_{sub.T}$, here expressed in %, with the limit temperature $T_{sub.0,amb}$ having been set here to 15° C. If the temperature of the supply air is below the limit temperature $T_{sub.0,amb}$, the correction factor can be defined for example as follows:

$$(33) F_T = \frac{T_{0,amb} - T_{amb,act}}{T_{0,amb} - T_{amb,act}} \cdot$$

(34) The correction factor for the stroke volume can be calculated according to the following formula:

$$(35) F_{Hub / KMK} = \frac{\text{currentstrokevolumeKMK}}{\text{maximumstrokevolumeKMK}} \cdot$$

(36) In a step **107**, the current cooling air volume flow $V_{sub.act}$ is compared with the setpoint for the cooling air volume flow defined in step **106**, and the difference between $V_{sub.soll}$ and $V_{sub.act}$ is formed, this difference defining a control deviation e . The fan **20** is then actuated in such a way that the cooling air volume flow $V_{sub.act}$ follows the specified setpoint for the cooling air volume flow $V_{sub.soll}$, namely in that, in step **108**, a current speed signal is transmitted to the fan motor **21** for the fan **20**. The method then begins again at step **103** or **105**, at a predefined sampling rate of, for example, 100 ms.

REFERENCE SIGNS

(37) **11**, **11'** screw compressor **12**, **12'** refrigeration dryer **13**, **13'** exhaust air duct **14**, **14'** compressed air line **15** dryer exhaust air duct **16** refrigeration dryer connection **17** compressor connection **18** cooling air outlet (screw compressor) **19** cooling air outlet (refrigeration dryer) **20**

fan 21 fan motor 22 controller 23 compressed air refrigeration heat exchanger 24 refrigerant circuit 25 refrigerant compressor 26 condenser 27 expansion valve 28 pressure sensor 29 exhaust air damper 30 flow sensor 31 inflow nozzle 39 drive motor 41 supply air inlet 42 ambient air sensor/temperature sensor 43 further compressed air line 44 consumer 45 screw compressor exhaust air duct 46 exhaust air damper (screw compressor) 47 branching point 48 first partial line 49 second partial line 50 third partial line 51 fourth partial line 52 joining point 53 further refrigeration dryer connection 54 cooling air outlet 55 dryer exhaust air duct 56 first output line 57 second output line 58 joining point 59 cooling air outlet 60 screw compressor exhaust air duct 61 compressor connection 62 compressed air inlet 63 compressed air outlet 64 exhaust air damper 65 cooling air flow 66 first output line 67 second output line 68 joining point 69 exhaust air damper

Claims

1. A compressed air station comprising at least two compressed air components that yield waste heat, wherein each compressed air component is designed either as a compressor or as a refrigeration dryer, wherein at least one of the at least two compressed air components, comprises a refrigeration dryer and is connected to an exhaust air duct, and wherein the other one of the at least two compressed air components is connected to the same exhaust air duct, wherein a compressed air refrigeration heat exchanger is provided within the refrigeration dryer, in which the compressed air is cooled by way of a refrigerant conducted in a refrigerant circuit, wherein the refrigerant circuit comprises a refrigerant compressor, a condenser, an expansion valve and the compressed air refrigeration heat exchanger, wherein the compressed air station further comprises a dryer exhaust air duct, which is provided for discharging a cooling air flow that is conducted through the refrigeration dryer, and which connects a cooling air outlet of the refrigeration dryer to a refrigeration dryer connection on the exhaust air duct, wherein the refrigeration dryer has a fan with a speed-adjustable fan motor, and the fan is designed to convey the cooling air flow even against a backpressure currently prevailing in the exhaust air duct, wherein the refrigeration dryer has a flow sensor for detecting a respective current value for the cooling air volume flow $V_{sub.act}$, and wherein the refrigeration dryer has a controller or interacts with a controller, which is configured and designed to record and process data from the flow sensor and to actuate the fan motor of the fan in such a way that, regardless of the current backpressure in the exhaust air duct, the respective current cooling air volume flow $V_{sub.act}$ follows a setpoint for the cooling air volume flow $V_{sub.soll}$.
2. The compressed air station according to claim 1, wherein the flow sensor is designed as a differential pressure sensor.
3. The compressed air station according to claim 2, wherein the flow sensor designed as a differential pressure sensor is assigned within an inflow nozzle assigned to the fan and detects the pressure prevailing there in comparison to a reference pressure.
4. The compressed air station according to claim 1, wherein the fan of the refrigeration dryer is designed as a radial fan.
5. The compressed air station according to claim 1, wherein a screw compressor and the refrigeration dryer are connected to one another via a compressed air line, and the compressed air line is designed to transfer the compressed air output by the screw compressor to the refrigeration dryer for drying purposes, wherein a cooling air outlet of the screw compressor is connected via a compressor connection to the exhaust air duct, to which the refrigeration dryer is also connected.
6. The compressed air station according to claim 1, wherein the setpoint for the cooling air volume flow $V_{sub.soll}$ is adapted as a function of a load state of the refrigerant compressor and/or as a function of the ambient temperature.
7. The compressed air station according to claim 1, wherein a setpoint for the cooling air volume flow $V_{sub.soll}$ results from a design value for the cooling air volume flow $V_{sub.nenn}$, which is

specific to the given refrigerant circuit with the given refrigerant compressor (25), according to the following formula:

$V_{\text{sub.soll}} = V_{\text{sub.nenn}} \cdot F_{\text{sub.Hub/KMK}} \cdot F_{\text{sub.T}}$, where the factors $F_{\text{sub.Hub/KMK}}$ and $F_{\text{sub.T}}$ are correction factors, each of which can alternatively be set to 1 or to a value between 0 and 1.

8. The compressed air station according to claim 1, wherein the controller interacts with an ambient air sensor for detecting a value representative of the state of the supply air and transmitting it to the controller.

9. The compressed air station according to claim 1, wherein an exhaust air damper is arranged in the dryer exhaust air duct, which damper is designed and configured to close the dryer exhaust air duct when the refrigerant compressor is idle.

10. The compressed air station according to claim 9, wherein the exhaust air damper is designed as a gravity-operated exhaust air damper, which opens whenever the fan is conveying the exhaust air through the dryer exhaust air duct and closes whenever the fan is idle.

11. The compressed air station according to claim 9, wherein the exhaust air damper interacts with a drive motor, wherein the drive motor is actuated by the controller in such a way that the exhaust air damper is opened or closed depending on the operating state of the fan.

12. A method for actuating a fan motor of a fan of a refrigeration dryer within a compressed air station, wherein the compressed air station comprises at least two compressed air components that yield waste heat, wherein each of the at least two compressed air component is designed either as a compressor or as a refrigeration dryer, and at least one exhaust air duct for discharging waste heat from a room, wherein at least one of the at least two compressed air components comprises the refrigeration dryer and is connected to the exhaust air duct, and wherein another one of the at least two compressed air components is connected to the same exhaust air duct, wherein a compressed air refrigeration heat exchanger is provided within the refrigeration dryer, in which the compressed air is cooled by way of a refrigerant conducted in a refrigerant circuit, wherein the refrigerant circuit comprises a refrigerant compressor, a condenser, an expansion valve and the compressed air refrigeration heat exchanger, wherein the compressed air station further comprises a dryer exhaust air duct, which is provided for discharging from the refrigeration dryer a cooling air flow that is conducted through the refrigeration dryer, and which connects a cooling air outlet of the refrigeration dryer to a refrigeration dryer connection on the exhaust air duct, wherein the refrigeration dryer has a fan with a speed-adjustable fan motor, and the fan is designed to convey the cooling air flow even against a backpressure currently prevailing in the exhaust air duct, wherein the method comprises: specifying a setpoint $V_{\text{sub.soll}}$ for the cooling air volume flow, detecting a respective current value for the cooling air volume flow $V_{\text{sub.act}}$, and actuating the fan motor of the fan in such a way that the respective current cooling air volume flow $V_{\text{sub.act}}$ follows the respectively specified setpoint for the cooling air volume flow $V_{\text{sub.soll}}$.

13. The method according to claim 12, wherein the respective current value for the cooling air volume flow $V_{\text{sub.act}}$ is detected by way of a differential pressure measurement.

14. The method according to claim 13, wherein in the differential pressure measurement a pressure is detected, relative to a reference pressure, immediately before the cooling air flow flows into the fan.

15. The method according to claim 13, wherein the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$ results from a design value for the cooling air volume flow $V_{\text{sub.nenn}}$, which is specified for the refrigerant circuit and the refrigerant compressors installed therein.

16. The method according to claim 13, wherein the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$ is adapted as a function of a load state of the refrigerant compressor and/or as a function of the ambient temperature.

17. The method according to claim 13, wherein the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$ results from a design value for the cooling air volume flow $V_{\text{sub.nenn}}$, which is specified for the refrigerant circuit and the refrigerant compressors installed therein, taking into

account one or more correction factors F.

18. The method according to claim 16, wherein the setpoint for the cooling air volume flow $V_{\text{sub.soll}}$ results from the design-based nominal value for the cooling air volume flow $V_{\text{sub.nenn}}$, according to the following formula:

$V_{\text{sub.soll}} = V_{\text{sub.nenn}} \cdot F_{\text{sub.Hub/KMK}} \cdot F_{\text{sub.T}}$, where $F_{\text{sub.Hub/KMK}}$ is a correction factor for taking into account the respective current stroke volume in the refrigerant circuit, and $F_{\text{sub.T}}$ is a correction factor for taking into account temperature fluctuations in the supply air of the cooling air flow, where:

$0 \leq F_{\text{sub.Hub/KMK}} \leq 1$ and $0 \leq F_{\text{sub.T}} \leq 1$.

19. The method according to claim 18, wherein the value T is set to 1 in the case of supply air temperatures above a limit temperature $T_{\text{sub.0,amb}}$, and:

$0 \leq F_{\text{sub.T}} < 1$ applies only in the case of cooling air inlet temperature values $T < T_{\text{sub.0,amb}}$.

20. The method according to claim 18, wherein the correction factor $F_{\text{sub.T}}$ for the cooling air inlet temperature T in ranges of the cooling air inlet temperature below the limit temperature $T_{\text{sub.0,amb}}$ is calculated as follows: $F_T = \frac{T}{T_{0, \text{amb}} - T_{\text{amb, act}}}$, where $\Delta_{\text{sub.T}}$ denotes a dryer-specific supplement in ° C., $T_{\text{sub.0,amb}}$ denotes a fixed limit temperature, and $T_{\text{sub.amb, act}}$ denotes the current supply air temperature.

21. The method according to claim 15, wherein the correction factor $F_{\text{sub.Hub/KMK}}$ is formed from a ratio of the current stroke volume and of the maximum stroke volume in the refrigerant circuit, according to the following formula $F_{\text{Hub / KMK}} = \frac{\text{currentstrokevolumeKMK}}{\text{maximumstrokevolumeKMK}}$.

22. The method according to claim 12, wherein a PID controller, a PI controller, a deadband controller or a three-point controller is used to actuate the fan motor of the fan (20) in order to bring $V_{\text{sub.act}}$ towards $V_{\text{sub.soll}}$, wherein a control deviation e results from $V_{\text{sub.soll}} - V_{\text{sub.act}}$, and wherein $V_{\text{sub.soll}}$ denotes the setpoint for the cooling air volume flow and $V_{\text{sub.act}}$ denotes the current cooling air volume flow.
